

Joshua Wrigley

Curriculum Vitae

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Appointments

2026 – 2028 **Masarykova Univerzita** – Brno, Czechia
Marie Skłodowska-Curie Action Postdoctoral Fellow,
Supervisor: [John Bourke](#).

The MSCA-PF is widely considered the most prestigious European postdoctoral fellowship. My project, [Higher Dimensional Aspects of First-Order Logic \(HiDimAFOL\)](#), received an evaluation score of 98.4% in a highly competitive application round – the 2025 round had a success rate of just 9.6% with a 64.6% increase in the number of applicants compared to the previous year.

Jan.–Feb. 2026 **Stellenbosch University** – Stellenbosch, South Africa
Visiting researcher.

Jan.–Dec. 2025 **Université Paris Cité** – Paris, France
Postdoctoral Researcher on project: [Topology for Types and Terms](#).
Principal investigator: [Sam van Gool](#).

During the position, I jointly supervised [Umberto Tarantino](#), a PhD student.

Apr.–Dec. 2024 **Queen Mary University of London** – London, UK
Postdoctoral Researcher,
Principal investigator: [Ivan Tomašić](#).

My position was supported by EPSRC Mathematical sciences small grant [Toward a Cohomology of Theories](#), which I jointly authored with Ivan Tomašić.

Grants

2026 – 2028 **Principal investigator:** [Higher Dimensional Aspects of First-Order Logic \(HiDimAFOL\)](#),
[Marie Skłodowska-Curie Action Postdoctoral fellowship](#) (€207,758).

2024 – 2025 **Research Co-Lead:** [Toward a Cohomology of Theories](#),
[EPSRC Mathematical sciences small grant](#) (£80,860),
Principal investigator: [Ivan Tomašić](#).

Education

2020 – 2024 **Università degli Studi dell'Insubria** – Como, Italy
PhD in Mathematics and Computer Science,
Thesis: [Doctrinal and Groupoidal Representations of Classifying Topoi](#), awarded *cum laude*,
Supervisor: [Olivia Caramello](#).

2019 – 2020 **Lincoln College, University of Oxford** – Oxford, UK
Masters in Mathematics (MMath), Classification: First class,
Dissertation: [Aspects of Stone Duality for Boolean Algebras](#),
Supervisor: [Hilary Priestley](#).

2016 – 2019 **Lincoln College, University of Oxford** – Oxford, UK
Bachelor of Mathematics (BA), Classification: First class.

Publications

2026 **Existential completions and Herbrand's theorem**,
Theory and Applications of Categories, Vol. 45, 2026, No. 23, pp 924-950 ([link](#)).

2026 **Day algebras**, with Edmund Robinson,
Mathematical Structures in Computer Science, Vol. 36, 2026, No. 6, pp 1–22 ([link](#)).

- 2026 **On Topological Groupoids that Represent Theories**,
Zeitschrift für Mathematische Logik und Grundlagen der Mathematik, to appear (2026) ([link](#)).
- 2025 **Topoi with enough points and topological groupoids**,
Journal of Pure and Applied Algebra, Vol. 229(10), 2025, 108073 ([link](#)).
- 2024 **Some Properties of Internal Locale Morphisms Externalised**,
Theory and Applications of Categories, Vol. 41, 2024, No. 35, pp 1160-1202 ([link](#)).
- 2024 **The Representing Localic Groupoid for a Geometric Theory**, with Graham Manuell,
Expositions in Theory and Applications of Categories, 2024, No. 2, pp 1-41 ([link](#)).

Preprints

- 2026 **Generic bundles over a localic category**, with Graham Manuell, [arXiv:2605.20407](#) (61 pages).
- 2025 **On the theories classified by an étendue**, [arXiv:2507.04526](#) (19 pages).
- 2025 **Ultracategories via Kan extensions of relative monads**, with Umberto Tarantino,
[arXiv:2506.09788](#) (41 pages).
- 2024 **Existentially closed models and locally zero-dimensional toposes**, with Mark Kamsma,
[arXiv:2406.02788](#) (30 pages).
- 2023 **The Geometric Completion of a Doctrine**, [arXiv:2304.07539](#) (63 pages).

Teaching experience

- 2025 **Graduate teaching** (Université Paris Cité)
Duality and categorical logic – 12hrs (course lecturer: [Sam van Gool](#)).
Lectured exercise classes, including assisting with course design and preparing problem sheets.
- 2022 – 2024 **Graduate teaching** (Università degli Studi dell’Insubria)
 - *Topics in category theory* – 36 hrs (course lecturer: [Olivia Caramello](#)),
 - *Topos theory* – 36 hrs (course lecturer: [Olivia Caramello](#)).*Lectured exercise classes and prepared corresponding lecture notes and problem sheets.*
- 2021 – 2022 **Undergraduate teaching** (Università degli Studi dell’Insubria)
 - *Algebra 1* – 12 hrs (course lecturer: [Valerio Monti](#)),
 - *Algebra 2* – 12 hrs (course lecturer: [Valerio Monti](#)),
 - *Geometria 1* – 12 hrs (course lecturer: [Olivia Caramello](#)).*Prepared and taught exam revision lessons, including preparation of problem sheets.
Taught in Italian.*

Invited lectures

- June 2026 **Co-ordinability in topos theory and model theory**,
[Workshop on Making Categorical Model Theory Accessible](#), Brno. Czechia.
- April 2026 **Towards a syntax-semantics description of non-commutative Stone duality via étendues**,
[7th Workshop on Formal Topology](#), Venice. Italy.
- Nov. 2025 **Day algebras**,
[Curry-Howard: Logic and Computation \(CHoCoLa\)](#), Lyon, France.
- Dec. 2024 **Theories, topological groupoids and topoi**,
[Categorical Logic and Higher Categories](#), Manchester, UK.

Professional activities

- June 2026 **Organiser** – [Ultracategories and Stone Duality Workshop](#), Paris, France.
- Dec. 2025 **Scientific committee** – [6th ItaCa Workshop](#), Milan, Italy.
- June 2025 **PhD examiner** – [Kristóf Kanalas](#) (Masaryk University, Czechia).
- 2024 **Undergraduate admissions interviewer** – Natural sciences, [St Edmund’s college](#), Cambridge.

Sept. 2022 **Local Organiser** – *9th Symposium on Compositional Structures*, Como, Italy.

Journal referee

Annals of Pure and Applied Logic; Theory and Applications of Categories; Applied Categorical Structures.